

Mobile Phone Sensing

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Mobile Phone Sensing System

- A system where sensor data is collected with the help of mobile phones and processed to allow the provision of useful information and services
- Emerging area of research and development
- Potentiated by the widespread proliferation of mobile devices

Classification

- Object of sensing
 - People-centered
 - Environment-centered
- Sensing domain
 - Home
 - Urban
 - Vehicular ...
- Types of mobile phone sensing
 - Individual
 - Participatory
 - Opportunistic
 - Crowd, social, etc.

Barriers to Adoption

- Social barriers
 - Privacy concerns
 - Restrictive laws in some countries
 - Absence of economic incentives that might encourage people to participate in a sensing campaign
- Technical barriers
 - Phone energy consumption
 - Variety of sensors and software for their management

Integrated Smartphone Sensors



- Microphone
 - Camera
 - GPS
 - Accelerometer
 - Compass (Magnetometer)
 - Gyroscope
 - Proximity
 - Light
-
- Barometric
 - Pressure (touch)
 - Others

External Sensors

- Other external sensor devices may be connected to the smartphone using a communication technology
 - Bluetooth
 - Bluetooth Low Energy
 - ANT
 - USB
- Examples of external sensors
 - Heart rate monitor
 - Posture sensors
 - Bicycle sensors (torque, cadence, power, etc.)

System Architecture



Applications

- **Individual sensing:**
 - data concerns mainly to the individual
 - examples: fitness applications, behaviour intervention
- **Group/community sensing:**
 - groups to sense common activities and **help achieving group goals**
 - examples: assessment of neighbourhood safety, environmental sensing, collective recycling efforts
- **Urban-scale sensing:**
 - large scale sensing, where large number of people have the same application installed
 - examples: tracking spread of disease across a city, traffic congestion, or pollution in a city

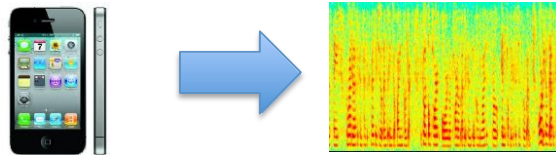
Nicholas D. Lane, Emiliano Miluzzo, Hong Lu, Daniel Peebles, Tanzeem Choudhury, Andrew T. Campbell. A Survey of Mobile Phone Sensing. IEEE Communications Magazine. September 2010

Mobile Phone Sensing Design

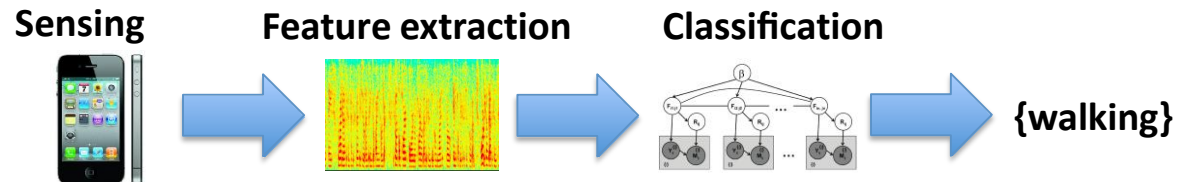
- Typical mobile phone sensing applications follow a common design pattern:
 - Collect raw data using the sensors of the mobile phone or environment.
 - Infer a particular activity of interest using the sensor values:
 - physical activity: is the user running?
 - context detection: is the user in a place full of other people?
 - Expose the high-level result to the user or use that result to adapt the behavior of the application.

Development Methodology

- Collect raw data



- Feature extraction and classification => Inference



- Mobile Sensing App Development
 - Extra features: storage, networking, sharing, privacy

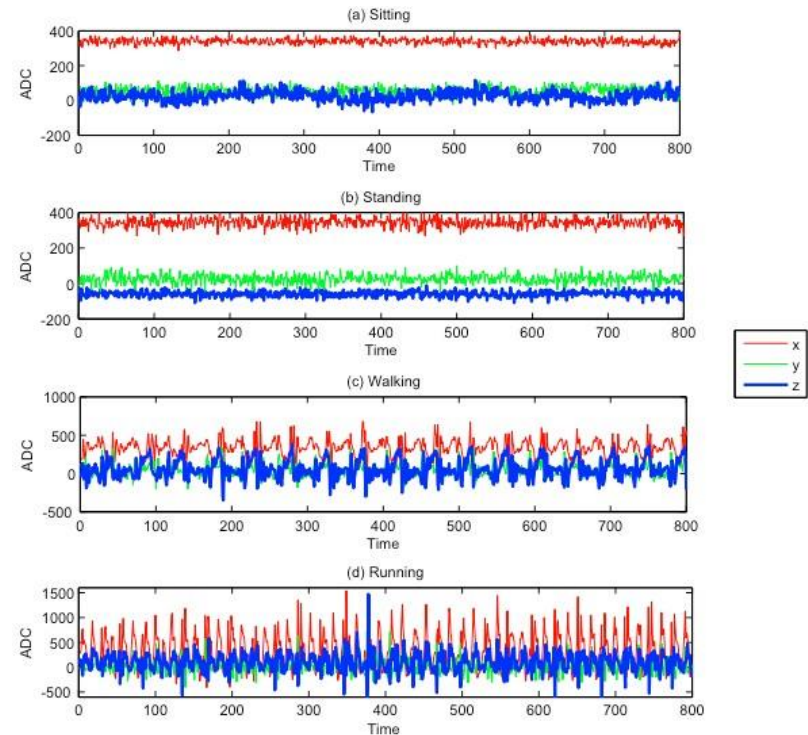


Feature Extraction

- Identifying features in a data set that can be used to infer a particular type of activity.
- The set of selected features depends on the type of sensor and the type of activity that is detected.
- The design process typically involves off-line analysis of training data to identify the right features for the particular inference engine.
 - Usually an iterative process where different features are tested.

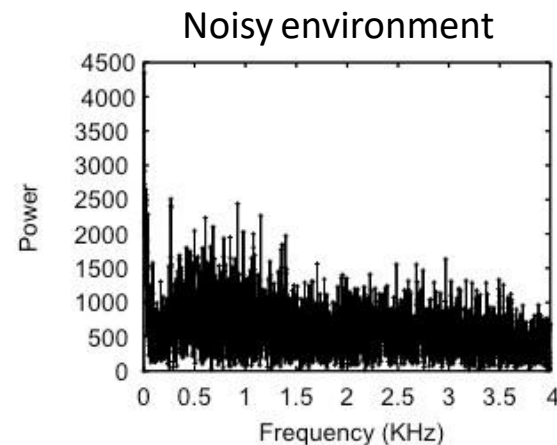
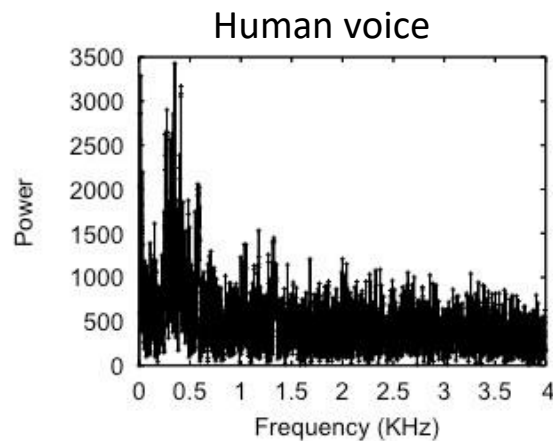
Example - Physical Activity Detection

- Sensor: accelerometer
- Activities: sitting, standing, waking, running
- Features:
 - Mean and standard deviation (can help distinguish between standing and walking).
 - Number of peaks per time (can help distinguish between walking and running).



Example - Conversation Detection

- Sensor - Microphone. Feature - FFT of sound
- Applying FFT on the audio samples, and comparing training data that are labelled as “conversation” and “non-conversation noise”.



- Sound samples of human voice present most of their energy within the 0-4 KHz spectrum.

Classification

- Feature extraction produces a feature vector (or set).
- The **classification matches the feature values** (feature vector) **to** a pre-defined set of high-level **classes** (e.g., **sitting, standing, walking, running**).
- The **classification** engine is typically **based on data mining/ machine-learning techniques** and is **trained previously using labelled training data**.
- Common classification algorithms include:
 - Naive Bayes classifier.
 - Decision Trees.
 - Hidden Markov Models.
 - Neural Networks.

Physical Activity

- Example Inferences:
 - {walking, running, up/down stairs}
- Sensors used:
 - Accelerometer, gyroscope, compass
- Applications:
 - Health/behaviour intervention
 - “Presence sharing”



Miluzzo et al. Sensing Meets Mobile Social Networks: The Design, Implementation and Evaluation of the CenceMe Application. Proceedings of SenSys'08.

Consolvo et al. Flowers or a Robot Army? Encouraging Awareness & Activity with Personal, Mobile Displays. Proceedings of UbiComp'08.

Transportation Mode

- Example Inferences:
 - {bike, bus, car}
- Sensors Used:
 - Accelerometer, GPS, Wi-Fi (location technologies)
- Applications:
 - Intelligent Transportation
 - Smart Commuting



Mun et al. PEIR, the personal environmental impact report, as a platform for participatory sensing systems research. Proceedings of MOBISYS '09

Context and Environment

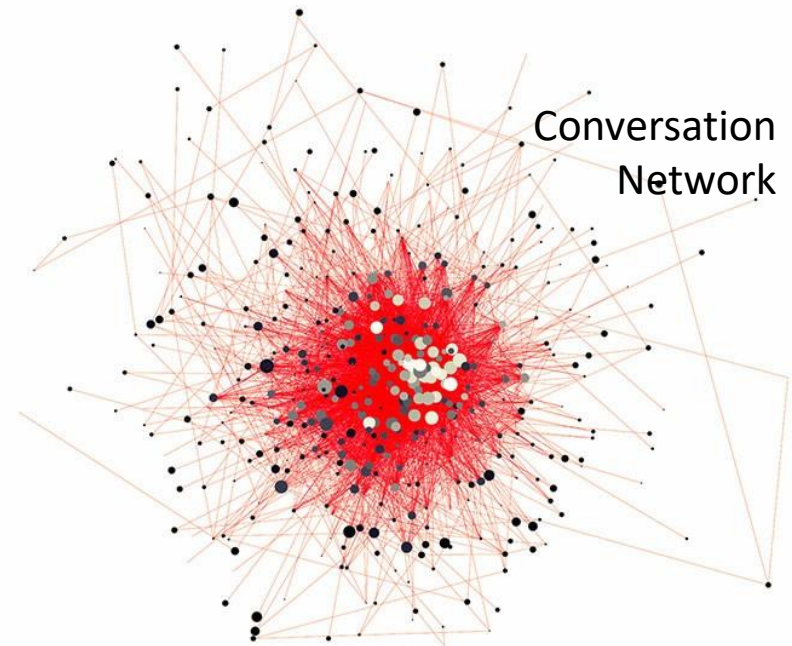
- Example Inferences:
 - {conversation, music, party, activity-related sounds}
- Sensors Used:
 - Microphone, camera
- Applications:
 - Automated Diary
 - Health & Wellness



Lu et al. SoundSense: scalable sound sensing for people-centric applications on mobile phones. Proceedings of MOBISYS '09

Human Voice and Conversations

- Example Inferences:
 - Stress, Speaker dominance
- Sensors Used:
 - Microphone
- Applications:
 - Social network analysis
 - Stress detection



Danny Wyatt, Tanzeem Choudhuri, Jeff Bilmes and James A. Ki's. Towards the Automated Social Analysis of Situated Speech Data. Proceedings of UbiComp'08

Detecting Emotions

- Example inference:
 - Emotional state, location and co-location with others
- Sensors used:
 - Microphone, Bluetooth, GPS
- Application:
 - Behaviour intervention
 - Computational social science
 - Using mobile sensing for quantifying theories in social science



Kiran Rachuri, Mirco Musolesi, Cecilia Mascolo, Peter J. Ren`row, Chris Longworth, and Andrius Aucinas. EmotionSense: A Mobile Phones based Adaptive Pla`orm for Experimental Social Psychology Research. Proceedings of UbiComp'10

Sensing

- Sensing is resource intensive



BATTERY



CPU



MEMORY

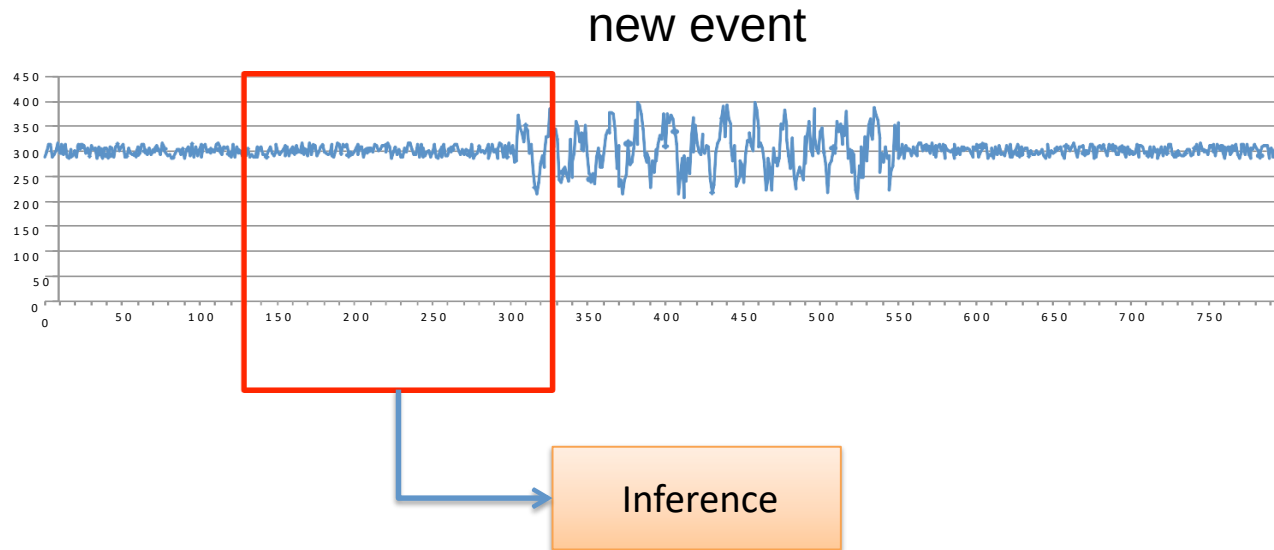


STORAGE

- The mobile phone's purpose is to support multiple applications.
- A mobile phone sensing application needs to maintain a balance between:
 - The amount of resources needed to operate.
 - The accuracy of the detection that is achieved.

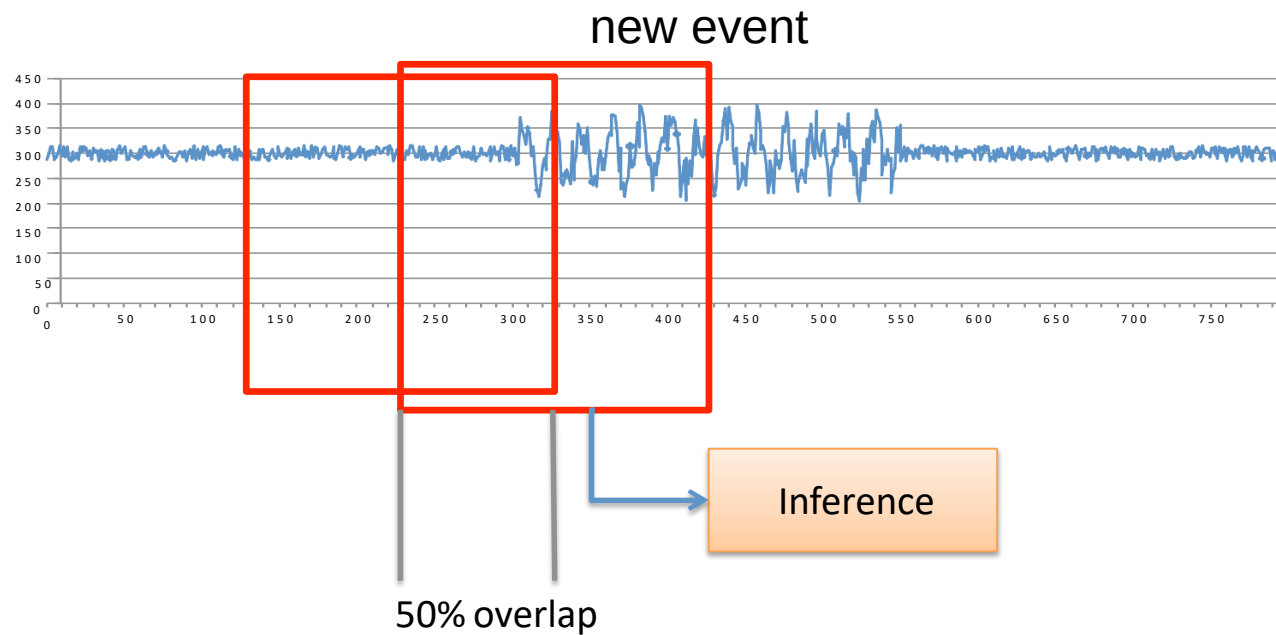
Sensing Modes

Continuous Sensing



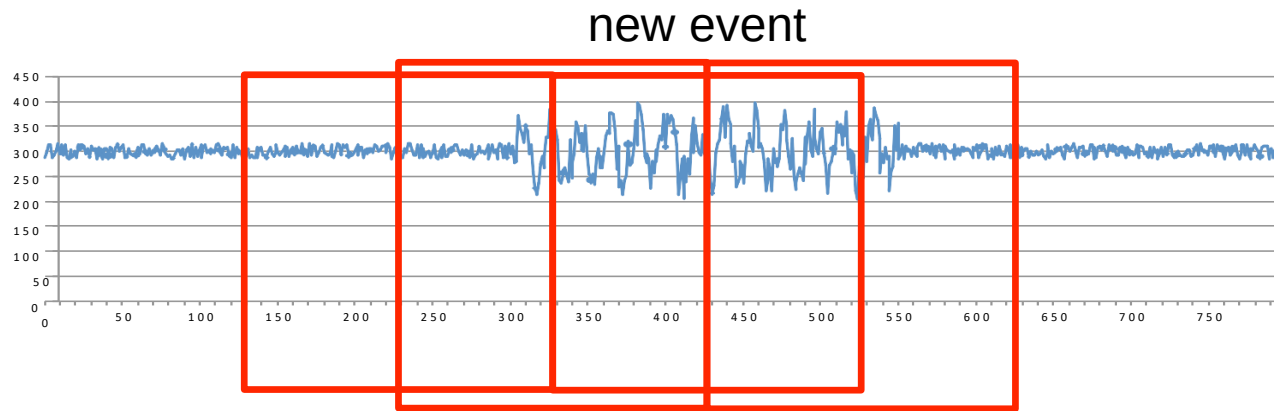
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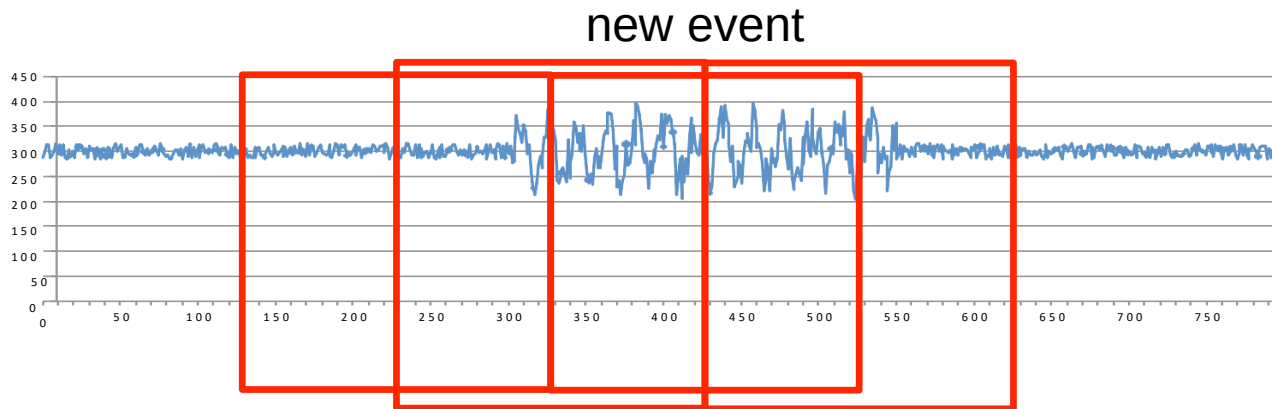
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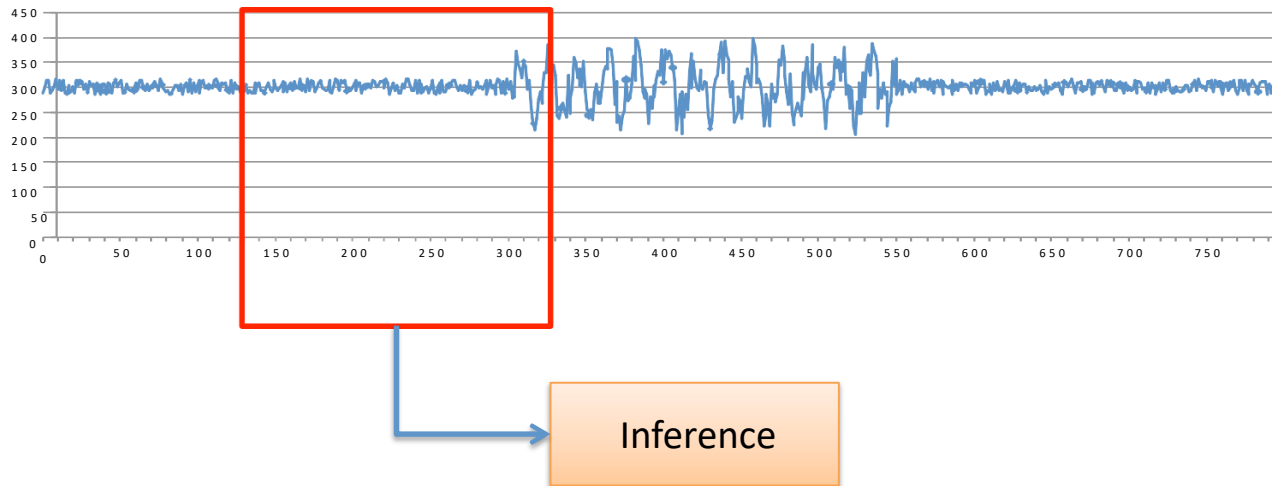
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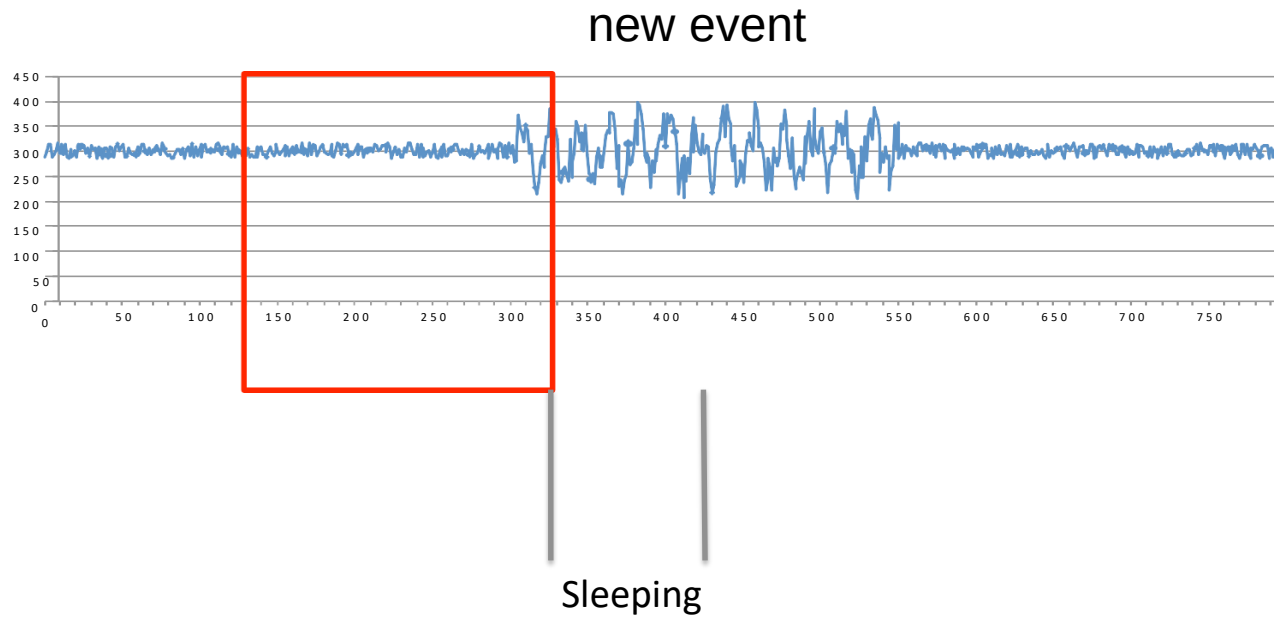


- Highly accurate data.
- Very costly in terms of battery and CPU usage:
 - Continuous sensing on multiple sensors can reduce phone standby to 6 hours.
 - May be used on “cheap” sensors, e.g. accelerometer.

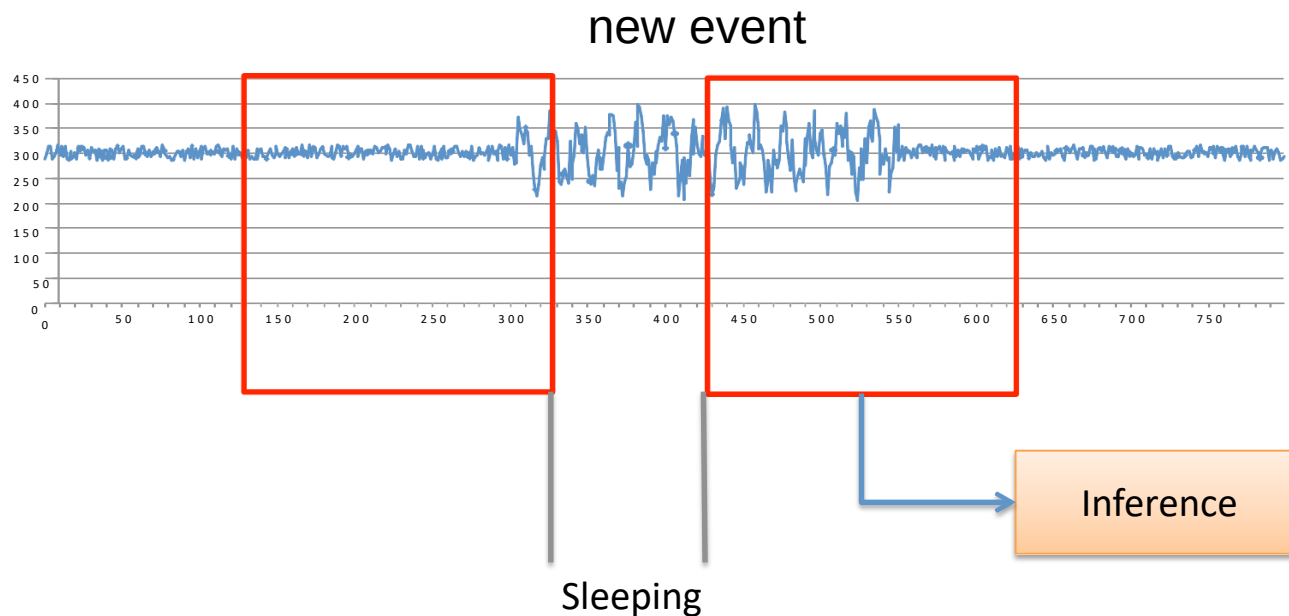
Sensing Mode Duty Cycling



Sensing Mode Duty Cycling



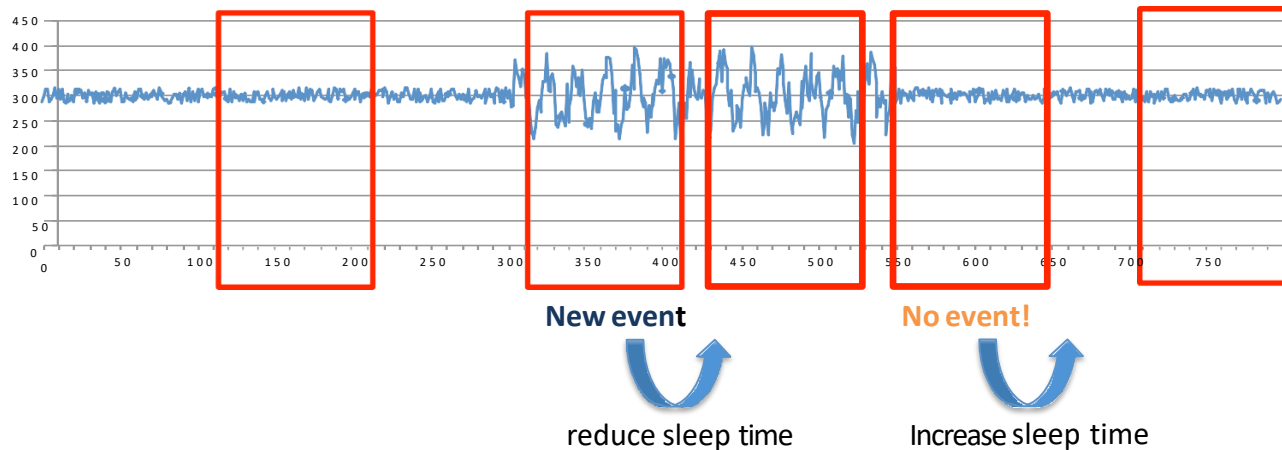
Sensing Mode Duty Cycling



- Lower impact on battery.
- Less accurate, interesting events may take place during sleeping.

Adaptive Duty Cycling

- Adjust the duration of sleeping periods according the rate of events that are detected.
- If no events are detected sleep for longer.
- When new events are detected reduce the sleeping time.



Adaptive Duty Cycling

- Typical approaches (depending on the type of events)
 - Exponential increase – linear decrease.
 - Linear increase – exponential decrease.
- Reduces the energy cost (compared to continuous sampling).
- Maintains high accuracy (compared to duty cycling).

But

- Requires a good understanding of the application domain:
 - in conversation detection a new voice events may be followed immediately by more such events, so faster sleep-time decrease may be necessary.
 - a location change event may not be followed by an immediate new event, so slower decrease may be applicable.

Bibliography

- Christos Efstratiou, Cecilia Mascolo, “Mobile Phone Sensing”, Mobile and Sensor Systems, Lecture 7, University of Cambridge, 2013.
- N. D. Lane, E. Miluzzo, H. Lu, D. Peebles, T. Choudhury, A. Campbell, “A survey of mobile phone sensing,” IEEE Computer Magazine, Vol. 48, No. 9, September 2010.
- Wazir Zada Khan et al. “Mobile phone sensing systems: A survey,” IEEE Communications Surveys & Tutorials, Vol. 15, No.1, 2013, pp. 402-427.
- Elsa Macias, Alvaro Suarez, Jaime Lloret, “Mobile Sensing Systems,” Sensors, Vol. 13, 2013.